

Autonome Eingebettete Systeme (AEiSy)

Level 6 – Kompassmodul

1. Kompassmodul – *CMPS14*
2. Kommunikationsschnittstelle (UART)
3. Aufgabe

Fakultät Elektrotechnik und Informatik

Level 6: Kompassmodul

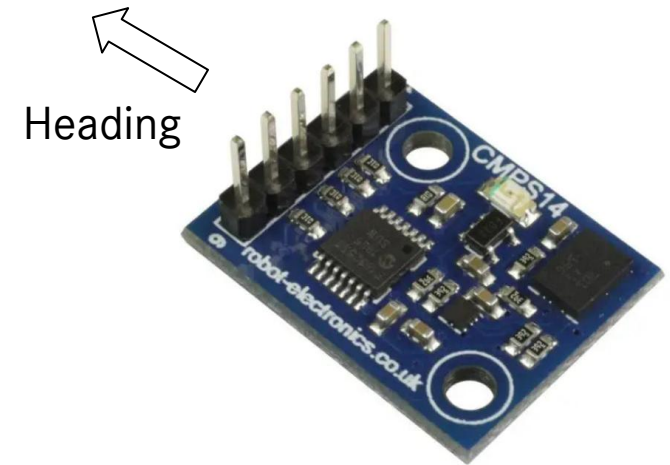
Lernziele:

- Inbetriebnahme des digitalen Kompasses
→ Bestimmung der Ausrichtung des Roboters
- Inbetriebnahme der notwendigen Peripherie des STM32G474 (UART)
- Design, Entwicklung und Test eines SW-Moduls in Abhängigkeit der Anforderungen und Randbedingungen (Ihre Lösungsstrategie!)
- Integration in bestehendes Embedded SW-System

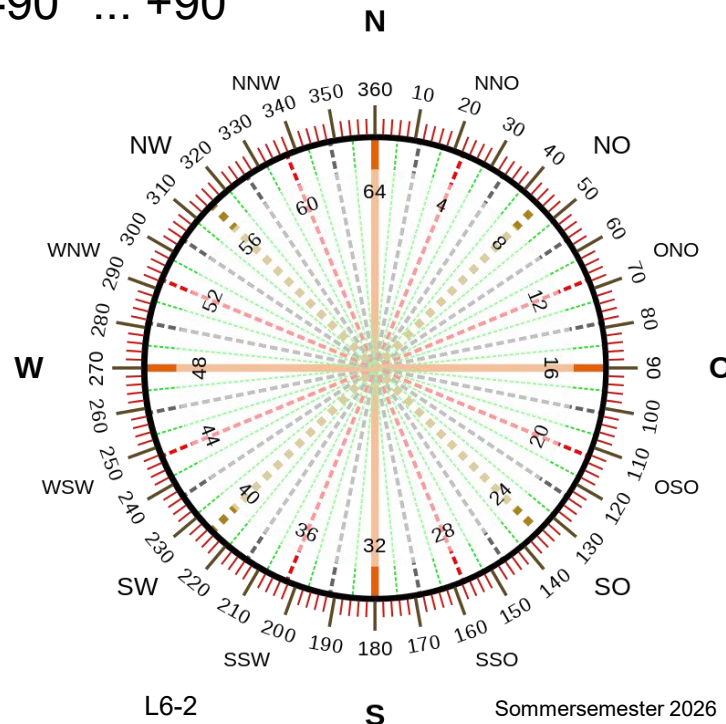
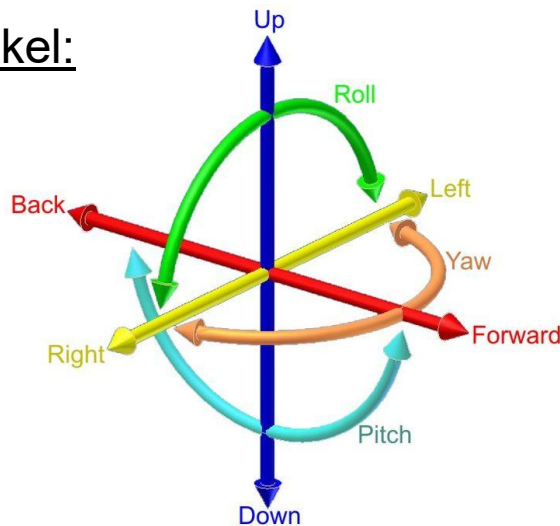
1. Kompassmodul – CMPS14

• Messwerte:

- Heading (16 Bit): 0 .. 3599 in 0,1°
- Heading (8 Bit): 0 .. 255
- Pitch: -90° ... +90°
- Roll: -90° ... +90°



• Winkel:



Heading:
Magnetischer Kompass,
intern korrigiert und
stabilisiert. Absolute
Orientierung zum
Nordpol.
Im Datenblatt auch als
Bearing bezeichnet.

2. Kommunikationsschnittstelle (UART)

- Schnittstelle zum μC :

→ *I2C / PWM* oder

→ **UART** (Universal Asynchronous Receiver Transmitter)

TxD PB10 (USART3_TX, AF7)

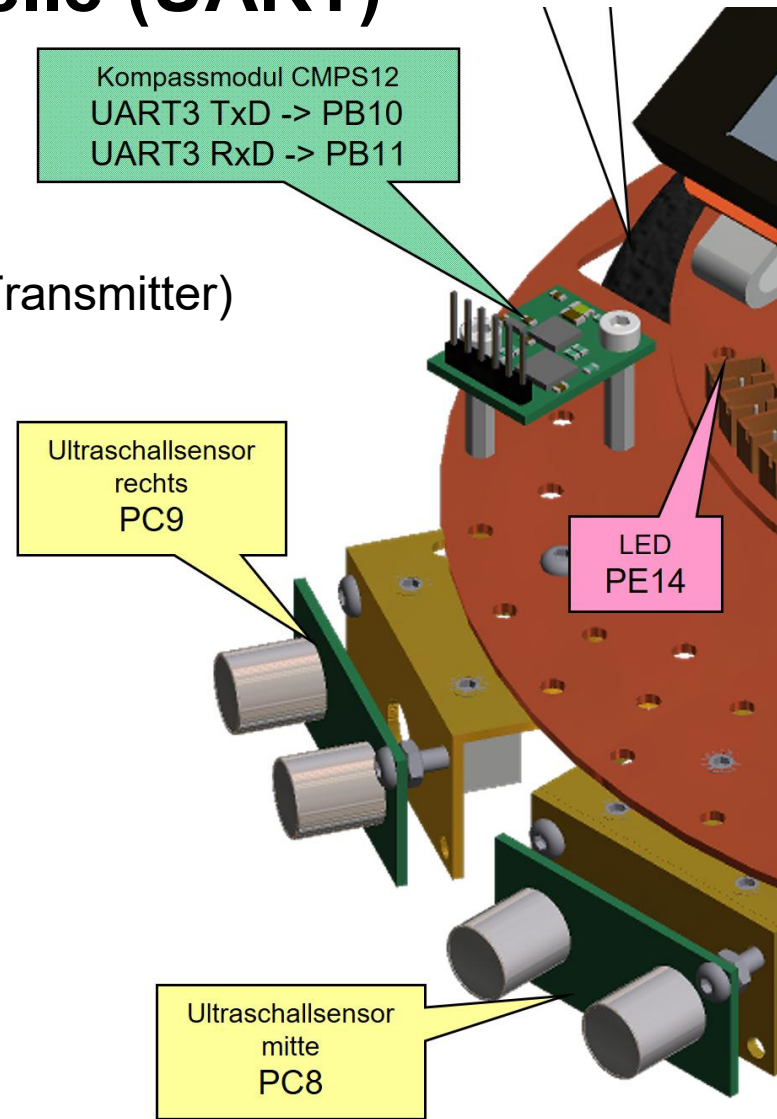
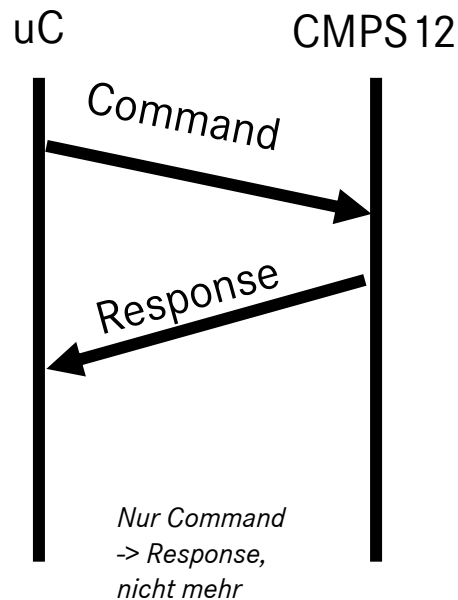
RxD PB11 (USART3_RX, AF7)

Baudrate = 9600

8 Datenbits

1 Stopp-Bit (!!)

Kein Paritätsbit



2. Kommunikationsschnittstelle (UART)

- Kommandos/
Messwerte:
(CMPS14)

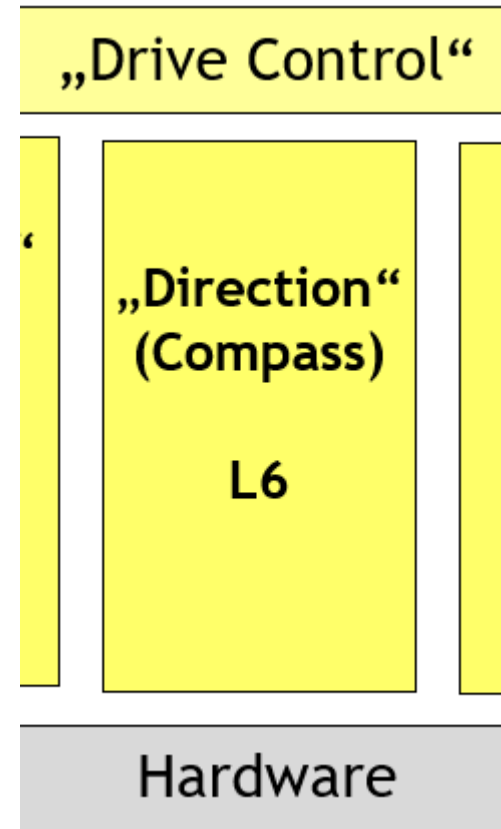
Command	Name	Bytes returned	Returned data description
0x11	GET VERSION	1	Software version
0x12	GET BEARING 8 BIT	1	Bearing as a single byte 0-255
0x13	GET BEARING 16 BIT	2	Bearing (16 bit), high byte first 0-3599
0x14	GET PITCH	1	Pitch angle +/- 0-90°
0x15	GET ROLL	1	Roll angle +/- 0-90°
0x19	GET MAG RAW	6	Raw magnetic data, 16 bit signed: X high, X low, Y high, Y low, Z high, Z low
0x20	GET ACCEL RAW	6	Raw accelerometer data, 16 bit signed: X high, X low, Y high, Y low, Z high, Z low
0x21	GET GYRO RAW	6	Raw gyro data, 16 bit signed: X high, X low, Y high, Y low, Z high, Z low
0x22	GET TEMP	2	BNO055 reported temperature as two bytes, high byte first and scaled in °C
0x23	GET ALL	4	Angle high, angle low (0-3599), pitch (+/- 0-90), roll (+/- 0-90)
			Bits 0 and 1 reflect the calibration status (0

- Updaterate: 100 Hz (s. Fusion output data rates, BNO080 Datenblatt)
10 Hz sollten ausreichend sein ...

3. Aufgaben

Moduldesign / Integration

- Definition der Schnittstellen zur Hardware (z.B. UART und GPIO) und übergeordneter SW-Schichten.
- Feindesign der Funktion „Compass“:
 - Definition der SW-Module (.c/.h Dateien) (z.B. uart.c/h / **cmps14.c/h**,...)
- Code-Entwicklung zur Messung der Winkel
 - Heading (Direction), Pitch, Roll
- Integration der entwickelten (und getesteten) SW-Module in die Gesamtarchitektur (→ Level 3)
- Darstellung der Ausrichtung/Heading auf dem LC-Display (→ Level 4)
- kontinuierliche Darstellung → Aktualisierung der LCD Anzeige



2. Kommunikationsschnittstelle (UART)

UART Basics:

Senden:

```
while (!(USART3->ISR & USART_ISR_TXE_TXFNF))  
    ;    // warten bis Transmit Data Register leer
```

Empfangen:

```
while (!(USART3->ISR & USART_ISR_RXNE_RXFNE))  
    ;    // warten bis Receive Data Register nicht leer
```

Blockierend: bei 3 Byte und 9600 Baud ca. 3 ms!

Anregungen:

DMA verwenden, RM0440 Kapitel 11 & 12.

Ist Polling bei 20Hz hinreichend, brauche ich Interrupts beim Empfangen?

„Geschicktes Polling“ *RXNE, ORE-Flag, ...*

Internetquellen für DMA und Interrupts.

(3) Einstellung der Kommunikationsparameter

→ *USART control register 1* (USART_x_CR1)

$x \in \{1, 2, 3, 4, 5\}$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	FIFO EN	M1	EOBIE	RTOIE	DEAT[4:0]					DEDT[4:0]				
		rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OVER8	CMIE	MME	M0	WAKE	PCE	PS	PEIE	TXEIE	TCIE	RXNEIE	IDLEIE	TE	RE	UESM	UE
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Reset Value: 0x00000000

← inkl. Paritätsbit !

Bit 28 **M1**: Word length

This bit must be used in conjunction with bit 12 (M0) to determine the word length. It is set or cleared by software.

M[1:0] = '00': 1 start bit, 8 Data bits, n Stop bit

M[1:0] = '01': 1 start bit, 9 Data bits, n Stop bit

M[1:0] = '10': 1 start bit, 7 Data bits, n Stop bit

This bit can only be written when the USART is disabled (UE=0).

Bit 10 **PCE**: Parity control enable

This bit selects the hardware parity control (generation and detection). When the parity control is enabled, the computed parity is inserted at the MSB position (9th bit if M=1; 8th bit if M=0) and the parity is checked on the received data. This bit is set and cleared by software. Once it is set, PCE is active after the current byte (in reception and in transmission).

0: Parity control disabled

1: Parity control enabled

This bitfield can only be written when the USART is disabled (UE=0).

Bit 0 **UE**: USART enable

When this bit is cleared, the USART prescalers and outputs are stopped immediately, and all current operations are discarded. The USART configuration is kept, but all the USART_ISR status flags are reset. This bit is set and cleared by software.

0: USART prescaler and outputs disabled, low-power mode

1: USART enabled

Bit 9 **PS**: Parity selection

This bit selects the odd or even parity when the parity generation/detection is enabled (PCE bit set). It is set and cleared by software. The parity is selected after the current byte.

0: Even parity

1: Odd parity

This bitfield can only be written when the USART is disabled (UE=0).

Bit 8 **PEIE**: PE interrupt enable

This bit is set and cleared by software.

0: Interrupt inhibited

1: USART interrupt generated whenever PE=1 in the USART_ISR register

Bit 3 **TE**: Transmitter enable

This bit enables the transmitter. It is set and cleared by software.

0: Transmitter is disabled

1: Transmitter is enabled

Bit 2 **RE**: Receiver enable

This bit enables the receiver. It is set and cleared by software.

0: Receiver is disabled

1: Receiver is enabled and begins searching for a start bit

→ *USART control register 2* (USART_x_CR2)

$x \in \{1, 2, 3, 4, 5\}$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
ADD[7:0]								RTOEN	ABRMOD[1:0]		ABREN	MSBFI RST	DATAINV	TXINV	RXINV
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SWAP	LINEN	STOP[1:0]		CLKEN	CPOL	CPHA	LBCL	Res.	LBDIE	LBDL	ADDM7	DIS_ NSS	Res.	Res.	SLVEN
rw	rw	rw	rw	rw	rw	rw	rw		rw	rw	rw	rw			rw

Reset Value: 0x00000000

Bits 13:12 **STOP[1:0]**: stop bits

These bits are used for programming the stop bits.

00: 1 stop bit

01: 0.5 stop bit.

10: 2 stop bits

11: 1.5 stop bits

This bitfield can only be written when the USART is disabled (UE=0).

→ *USART control register 3* (USART_x_CR3) → für uns nicht relevant !